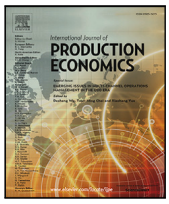
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win–win contract farming in dual-channel agribusiness supply chains under yield, quality, and price uncertainty

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ABSTRACT

Despite the benefits of contract farming, power imbalances between farmers and agribusiness firms often result in unfair agreements. This study aims to design a fair (win–win) contract farming model for a dual-channel (fresh and processed) blueberry supply chain, incorporating an incentive mechanism to ensure that all parties benefit. We examine the benefits of incentive-based contracts over penalty-based contracts by analytically investigating three farming situations: (1) no contract, (2) penalty-based contract farming, and (3) incentive-based contract farming. We establish analytical conditions for collaborative incentive-based contract terms that benefit both parties and validate these conditions numerically using data from a representative blueberry farm. Our findings indicate that the incentive-based contract farming leads to mutually beneficial outcomes and higher supply chain profits compared to the penalty-based contract. We conduct numerical and comprehensive sensitivity analyses to assess the impact of contract farming on the profits of the farmer, agribusiness firm, and the overall supply chain. Our study has significant theoretical and practical implications, emphasizing the importance of balanced and mutually beneficial contract farming arrangements that account for yield, price, and quality uncertainties within a dual-channel supply chain.

1. Introduction

Contract farming, a prevalent mechanism within agricultural supply chain management, represents a strategic relationship between agribusiness firms (distributors) and agricultural producers (farmers). This legal arrangement facilitates the production and distribution of specific agricultural commodities through agreements that define product quantity, quality, and pricing. While these arrangements provide critical market access for farmers and ensure a consistent supply for distributors, they encapsulate complex dynamics such as resource allocation, risk management, and power asymmetries that necessitate a sophisticated operational research approach to optimize and balance the interests involved (Federgruen et al., 2019; Mishra et al., 2018; Chen and Chen, 2021).

The literature delineates two predominant forms of contract farming: marketing and production contracts. Marketing contracts obligate the farmer to produce a specified quantity at a predetermined price, focusing on stabilizing revenue streams for farmers. Production contracts extend further the power of the buyer into the farmer's production decisions, often by providing inputs and then aligning the production quantities and quality with the distributor' supply chain management practices (Federgruen et al., 2019). While these contractual arrangements can effectively mitigate operational risks such as price and demand uncertainties, critiques highlight that they may inadvertently perpetuate or exacerbate existing power imbalances, potentially constraining the operational flexibility and economic returns of farmers (Warning and Key, 2002; Narayanan, 2014; Mwambi et al., 2016). Therefore, we propose to examine penalty-based and incentive-based contracts to address the power imbalance between farmers and distributors.

Our study addresses these critical issues within the blueberry market in British Columbia (BC), a sector characterized by significant economic value and susceptibility to environmental and market volatility (IndexBox, 2022). BC is a major blueberry producer, making it a well-suited region for this study. Its well-developed blueberry industry operates within a dual-channel supply chain (fresh and processed), providing valuable insights into the complexities of supply chain management, particularly in relation to yield variability and

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market fluctuations. Furthermore, BC's climatic conditions and farming practices reflect key challenges faced by blueberry producers globally, including climate sensitivity and yield uncertainty. The integration of BC-based data enhances the credibility of the model, while the analytical methodology remains adaptable to other regions or crops through parameter adjustments that account for local conditions and supply chain structures.

Blueberries, unlike many annual crops, are long-lived perennial shrubs that require several years to mature and can remain productive for decades under optimal conditions (Strik and Yarborough, 2005). While most high-bush blueberry plants reach full production within 6 to 8 years, well-managed bushes can continue bearing fruit for over 20 years, with some reports indicating productivity up to 50 years (Strik, 2015). This extended production cycle presents distinct investment and management challenges compared to short-term crops such as tomatoes, cucumbers, or lettuce (Ballington, 2009).

Blueberry cultivation is highly sensitive to soil conditions, climate, and environmental stressors (Rowland et al., 2020; Saftner et al., 2008; Bryla and Strik, 2007). These factors contribute to significant uncertainty in both market supply and pricing, necessitating tailored decision-making models to optimize production and supply chain strategies. Nonetheless, our methodological framework employed in this study remains adaptable to similar crop supply chains through parameter calibration, particularly in addressing uncertainties related to yield, quality, and market conditions (Evans and Ballen, 2019).

We examine penalty-based and incentive-based contracts against a no-contract baseline in a dual-channel blueberry supply chain. We employ a Stackelberg game approach to model the hierarchical decision-making processes inherent in these contracts, where the distributor typically assumes the leadership role (Huh and Lall, 2013). Penalty-based contracts, traditionally favoured in agri-food supply chain contracts, impose financial penalties on farmers for non-compliance with the supply agreements, effectively transferring a significant portion of operational risk to farmers. Such contracts align with a risk-averse supply chain strategy from the distributor's perspective but may not incentivize optimal production behaviour by the farmer. Penalties in contract farming agreements may be imposed for various reasons, including non-compliance with quality standards, delays in delivery, or failure to meet the contracted quantity. These enforcement mechanisms are intended to mitigate risks for buyers and ensure adherence to contractual terms. The Legal Guide on Contract Farming, developed by The International Institute for the Unification of Private Law (UNIDROIT, 2015), provides a comprehensive analysis of the legal framework governing such agreements. It outlines the rights and obligations of contracting parties and details the potential consequences of non-compliance, which often include financial penalties and other contractual sanctions aimed at safeguarding supply chain stability.

Conversely, incentive-based contracts represent a novel contribution to the field by embedding mechanisms that align the incentives of both farmers and distributors. We propose contracts incorporating production cost-sharing and risk-pooling strategies, which have been recognized in operational research for enhancing supply chain resilience and flexibility (Osei-Bryson and Ngwenyama, 2006; Bakshi and Kleindorfer, 2009; Dong et al., 2012). By sharing the risks associated with yield variability and price fluctuations, these contracts promote a more collaborative approach to supply chain management, encouraging both parties to follow practices that enhance overall performance.

Our research investigates and validates the efficacy of incentive-based contract farming. Our findings indicate that incentive-based contracts create a more balanced collaborative relationships and enhance the overall supply chain surplus, providing compelling evidence for rethinking traditional contract farming arrangements. This study contributes to academic discourse and offers substantial managerial implications, suggesting that more equitable and efficient contract designs can be developed by integrating incentive mechanisms such as production cost sharing.

The remainder of this study is structured as follows. Section 2 reviews existing literature to position our research within the ongoing scholarly debate. Section 3 defines the problem and introduces the notations and parameters utilized in our analysis. Section 4 provides an analytical comparison of penalty- and incentive-based contracts using no-contract situation as a baseline. Section 5 discusses the results from our numerical experiments, providing numerical support for our theoretical models. Section 6 translates our findings into actionable managerial insights that can be applied to enhance contract farming practices. Finally, Section 7 summarizes the contributions of our research and outlines potential avenues for future exploration in this area.

2. Literature review

Our study contributes to the existing literature on agri-food supply chain contracting, specifically contract farming. Eaton and Shepherd (2001) defines contract farming as agreements between farmers and agri-food firms for the future supply and production of agricultural products at predetermined prices.

A futures contract is a binding agreement where the buyer must purchase an asset or commodity at a predetermined price on a specified future date (Prager et al., 2020). Farmers use futures contracts to hedge against price risks in agriculture, securing a guaranteed price before the selling season. Previous research by Paul (1976) examined the pros and cons of forward and futures contracts in agriculture, while Tashjian (1995) discussed theoretical features of effective futures contracts. Brorsen and Fofana (2001) identified cash market activeness as a primary determinant of futures contract success.

Studies have explored futures contract in specific markets. Yuan et al. (2007) modelled supply functions with futures contracts in different bidding strategies in the tea market. Mathews (2008) proposed a sustainably certifiable bio-fuels futures contract for a global market. Pinçe et al. (2021) developed a response framework for a commercial seed supply chain under contractual arrangements. Additionally, Chen et al. (2019) and Liu and Wang (2019) examined options within futures contracts, focusing on time-dependent spot price volatility and financial risk hedging in supply chains, respectively.

Extensive research has identified factors contributing to the success of futures contracts across various markets. For instance, Fung and Liu (2013) found that price-discovery improvements are significant in China's stock market. Oliveira et al. (2013) demonstrated that futures contracts in a two-level electricity supply chain lead to lower retail prices, efficiency improvements, and reduced double marginalization. de Zegher et al. (2019) highlighted that contracts can create shared value in wool supply chains, while Qian and Olsen (2022) emphasized their relevance for quality coordination in agricultural cooperatives.

Agricultural supply chains face challenges in managing multi-tier partnerships, particularly in dealing with complex contractual relationships and a lack of transparency in transactions (Kamble et al., 2020). Various types of futures contracts and agreements are commonly used in agricultural supply chains, including bonus and penalty contracts (Burer et al., 2008), cost-sharing (Niu et al., 2016), partially guaranteed-price contracts (Tang et al., 2016), and options contracts (Yang et al., 2017). Anderson and Monjardino (2019) proposed double discount contracts and cost-sharing/buyback contracts for coordinating supplier–grower–buyer chains. Coordination mechanisms using options contracts and dynamic incentives for risk transfer in agricultural supply chains were developed by Zhang et al. (2023). Zhong et al. (2023a) developed a game-theoretical model to examine the impact of extreme weather risks on agricultural supply chains under full-cost and income insurance schemes. Optimal forward contract design and multi-period crop planning under uncertainty have also been studied (Chen and Ryan, 2023), along with the effects of demand information sharing (Hong et al., 2023) and ambiguity aversion (Zhong

et al., 2023b). Contracts were also found to create shared value in wool supply chains (de Zegher et al., 2019) and to be relevant for quality coordination in agricultural cooperatives (Qian and Olsen, 2022).

The incorporation of penalty clauses in supply contracts is a common practice (Mathur and Shah, 2008). The literature on contract farming has also predominantly concentrated on penalty-based mechanisms or production control mechanisms, where farmers incur penalties for not meeting agreed-upon quantities or give away their agency. This focus has spurred debate on the efficacy and fairness of such penalties or controls. Abebe et al. (2013) studied the motivations of smallholder potato farmers in Ethiopia, finding that punitive measures for exceeding size specifications led farmers to alternative markets to avoid penalties, indicating a misalignment between farmer incentives and contractual obligations.

Huh and Lall (2013) demonstrated that farmers' preference for contract farming over independent farming hinges on contract details, particularly prices exceeding expected market prices to compensate for penalties. Peng and Pang (2019) analysed a three-tier contract farming supply chain, considering government subsidies, yield uncertainty, and the risk-averse behaviour of farmers. Using game-theoretic models, Shi and Cao (2020) explored coalition formation among agricultural producers to manage yield uncertainty, concluding that higher penalties reduce profits and discourage cooperative behaviour. Rajput and Venkataraman (2022) introduced a model resilient to extreme market price fluctuations, finding that higher penalties lead to reduced land allocation for contract crops.

While previous studies have primarily examined the role of penalties in contract farming, comparative analyses of incentive-based and penalty-based contracts within agricultural supply chains remain limited. Although economic and behavioural theories suggest that incentive-based mechanisms can be more effective than punitive measures in promoting compliance and performance (Lee et al., 2018; Cai et al., 2023), empirical and analytical research demonstrating these advantages in agricultural contract farming is scarce. This gap underscores the need for further investigation into how incentive structures can improve supply chain efficiency and promote more equitable relationships between farmers and buyers.

A notable exception in the literature is Saenger et al. (2013), which compared penalties and incentives in the Vietnamese dairy sector. Their empirical analysis showed that while penalties encouraged greater input use, bonus payments further enhanced this effect, though they also highlighted the economic challenge of balancing costs and benefits. Unlike Saenger et al. (2013), our study is the first to analytically compare penalty-based and incentive-based frameworks in contract farming and design an incentive-based contract that benefits farmers and distributors.

Contract farming has the potential to benefit farmers and distributors in agricultural supply chains. However, due to power imbalances between the parties and uncertain nature of the farming business, these contracts may only sometimes result in mutually beneficial outcomes. This study aims to analytically identify the conditions necessary for designing a contract farming system that is fair and mutually beneficial under various uncertainties. By shifting the focus from punitive to positive reinforcement mechanisms, our research seeks to demonstrate how incentive-based strategies can enhance the attractiveness of contract farming for both parties, contributing to a more balanced perspective on contract terms in the agricultural sector.

3. Problem definition

We introduce the notations used throughout this analysis in Table 1. We focus on a dual-channel (fresh and processed) blueberry supply chain that includes options for fresh spot market sales, fresh agri-food distribution, and processed distribution channel, as depicted in Fig. 1. During the cultivation season, farmers must strategically decide how to allocate their harvest to the fresh or processed markets.

To formulate the problem, we consider a single representative farmer as a proxy for a collective of farmers (e.g., the BC Blueberry Council). This approach provides a practical abstraction that captures the average performance and strategic decision-making of multiple farmers under a unified framework. By modelling the collective as a single meta-entity, the analysis accounts for coordinated supply chain decisions. However, in contexts where individual farmers make independent choices, factors such as market competition, price differentiation, and strategic interactions would need to be incorporated to more accurately reflect the complexities of a decentralized farming system.

The fresh and processed blueberry channels differ in price stability, infrastructure needs, and quality requirements. Fresh blueberries command higher but more volatile prices, requiring careful planning, specialized storage, and rapid distribution. In contrast, processed blueberries offer a lower but stable price and require less stringent handling. Quality standards also vary, with fresh blueberries should meet strict size and firmness criteria, while processed blueberries face fewer quality restrictions. Despite these differences, farmers must strategically allocate their harvest between the two channels. As perennial crops, blueberries allow for long-term planning, though short-term allocation depends on harvest conditions and market demand.

To qualify for the fresh market, blueberries must meet stringent quality standards such as ripeness, size, colour, firmness, and minimal fruit rot and insect pest contamination. These quality grades are significantly influenced by various cultivation and environmental factors (Rowland et al., 2020; Cortés-Rojas et al., 2016). Blueberry size is widely recognized as a key quality attribute influencing market acceptance and consumer preference in the fresh blueberry sector (Donahue et al., 2000; Saftner et al., 2008; Johnson et al., 2011; Jorquera-Fontena et al., 2017; Milić et al., 2018). Larger berries are generally favoured due to their superior texture, flavour, and ease of consumption, making size a critical factor in pricing and grading standards. Market trends further reinforce the importance of size as a practical quality indicator. Observations from the BC blueberry market and international markets indicate a consistent consumer preference for larger berries, which strongly influences pricing, retail positioning, and purchasing behaviour.

In our model, the diameter of a blueberry serves as the main proxy for quality, encapsulated by an uncertain parameter θ_f described by a probability density function (PDF) $h(\theta_f)$ and a cumulative distribution function (CDF) $H(\theta_f)$. The proportion of the harvest that qualifies as fresh marketable blueberries is $1 - H(\Theta_f)$, where Θ_f represents the quality threshold. This calculation estimates the portion of the total realized yield that meets the fresh market standards and has the potential to be sold as fresh blueberries.

Farmers have two avenues for selling their fresh products: directly on the spot market or through contracts with agri-food distributors. The spot market includes diverse selling points such as farmers' markets, local retail stores, e-commerce platforms, on-farm stores, roadside stands, and community-supported agriculture programmes. Due to the high perishability of blueberries, selling on the spot market necessitates in advance planning and effective logistics to ensure feasibility. Consequently, decisions to engage in direct sales via the spot market must be made well in advance, as last-minute diversions of surplus blueberries to these outlets are logistically impractical.

Alternatively, farmers may choose to collaborate with established agri-food distributors. These distributors maintain a strong market presence and operate extensive and efficient distribution networks encompassing large retail chains, restaurants, and other commercial buyers. They are equipped with the necessary logistics and infrastructure to transport blueberries across extensive geographical areas, both nationally and internationally, thereby facilitating the movement of fresh produce through more comprehensive and complex market channels.

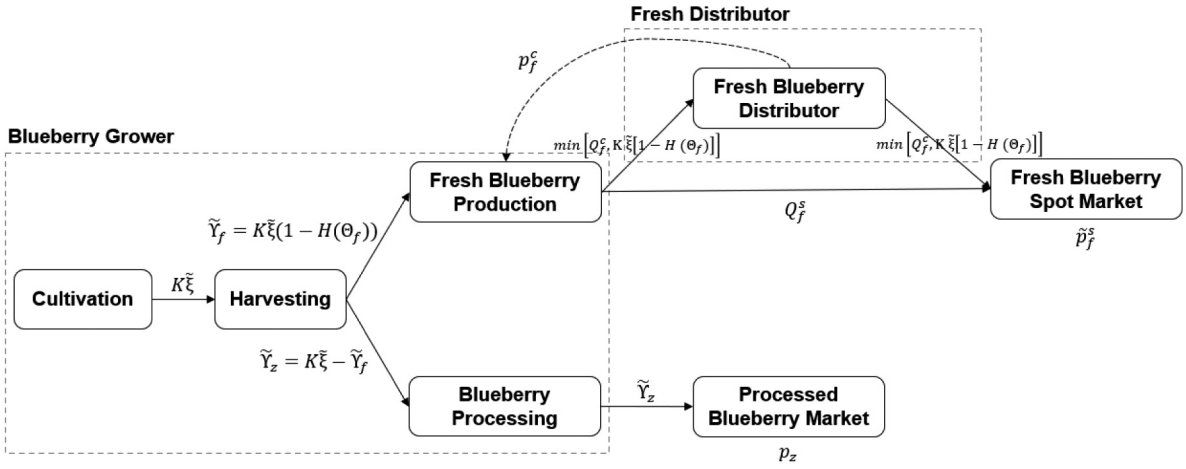


Fig. 1. The structure of the investigated supply chain.

Table 1
Notations.

Parameters	
K	Size of the blueberry farm (in acres or hectares)
p_z	Unit wholesale price of processed blueberries
c_f	Unit variable cost of harvesting fresh blueberries
c_z	Unit variable cost of processing blueberries
C_s	Fixed distribution cost for the farmer to directly distribute fresh blueberries to the spot market
C_d	Fixed distribution cost for the distributor
c_d	Unit variable cost of distributing blueberries for the distributor
c_s	Unit variable cost of distributing blueberries for the farmer
$\tilde{\xi}$	Random variable representing the uncertain yield rate of blueberry production, with cumulative distribution function $G(\xi)$ and probability density function $g(\xi)$
$\tilde{p}_f^s$	Random variable representing the uncertain spot market price of fresh blueberries, with cumulative distribution function $F(p_f^s)$ and probability density function $f(p_f^s)$
$\tilde{\theta}_f$	Random variable representing the uncertain quality of blueberries, with cumulative distribution function $H(\theta_f)$ and probability density function $h(\theta_f)$
Θ_f	Minimum blueberry diameter required to meet fresh market quality standards
π	Penalty per unit of unfulfilled fresh blueberries imposed by the distributor, expressed as a percentage of the wholesale price
$\tilde{Y}_f$	Realized yield of fresh blueberries that meet the quality standards
$\tilde{Y}_z$	Realized yield of blueberries to be sold as processed
Decision variables	
p_f^c	Guaranteed price offered by the distributor for fresh blueberries under penalty-based contract farming
p_f^{sc}	Guaranteed price offered by the distributor for fresh blueberries under incentive-based contract farming
Q_f^s	Quantity of fresh blueberries to be sold directly by the farmer on the spot market
Q_f^c	Committed quantity of fresh blueberries by the farmer under penalty-based contract farming
Q_f^{sc}	Committed quantity of fresh blueberries by the farmer under incentive-based contract farming
α^{sc}	Proportion of production unit cost subsidized by the distributor under incentive-based contract farming

Processing excess blueberries through shared-use facilities or third-party processors can be a beneficial strategy for managing surplus production. While processed blueberries offer stable prices and may have lower profit margins than fresh ones, they provide a reliable outlet for surplus production, ensuring that no harvest goes to waste.

Farmers and distributors frequently enter contractual agreements establishing a guaranteed wholesale price and providing financial stability for farmers. Unlike processed blueberries, which can be stored and sold later, fresh blueberries have a limited shelf life, requiring rapid sales to prevent spoilage. Therefore, contract farming serves as a practical mechanism in the fresh blueberry market, offering farmers a stable sales channel and mitigating market uncertainties.

Fig. 2 outlines the timeline for contract farming in the blueberry supply chain, where negotiations between farmers and distributors typically occur in late spring or early summer during the cultivation season. In these contracts, the distributor commits to purchasing a specified quantity of blueberries from the farmer at a guaranteed wholesale price p_f^c, contingent on the blueberries meeting the quality standards for fresh produce. The farmer determines the production quantity Q_f^c based on this agreed-upon wholesale price (Hong et al., 2023). This model of contract farming is consistent with the existing body of literature, including seminal works by Niu et al. (2016), Nematollahi et al. (2021), and Hong et al. (2023), which explore similar strategies and their implications.

As illustrated in Fig. 2, the blueberry varieties Elliott, Bluecrop, and Duke exhibit distinct harvest timing, ripening patterns, and market advantages (Bryla and Strik, 2007; Strik et al., 2017). Duke, an early-season variety, allows growers to enter the market sooner, though achieving optimal yields often requires multiple harvests. Bluecrop, a mid-season variety, is valued for its uniform ripening, which enhances harvesting efficiency and fruit quality consistency. Elliott, a late-season variety, extends the availability of fresh blueberries and performs well in cooler climates. These differences influence market timing and cultivation strategies, aligning with growers' economic and logistical considerations. However, this study focuses on the comparison of penalty-based and incentive-based contract farming within dual-channel agribusiness supply chains. To maintain analytical clarity, varietal differences are not incorporated into the problem formulation; instead, blueberries are categorized into two broad market segments: fresh and processed, aligning with the study's objectives.

A prevalent method in operations management for addressing unpredictability in production environments is to link production costs directly to pre-planned quantities (He and Zhang, 2008; Yuan et al., 2021; Xie et al., 2021). This approach is particularly relevant to blueberry cultivation, where early preparations are crucial. It is based on the understanding that production costs depend on planned quantities before the harvesting season's onset.

Additionally, the farm's productive capacity is influenced by factors such as farm size K and the uncertain yield $\tilde{\xi}$, characterized by a cumulative distribution function (CDF) $G(\xi)$ and a probability density function (PDF) $g(\xi)$. The pricing dynamics for blueberries are also complex; fresh blueberries face price fluctuations due to a globally competitive market and perishability, impacting the farmer–distributor relationship significantly, whereas processed blueberries have stable prices throughout the season (USHBC, 2020). While the seasonality of fresh blueberry market prices is well recognized, this study focuses

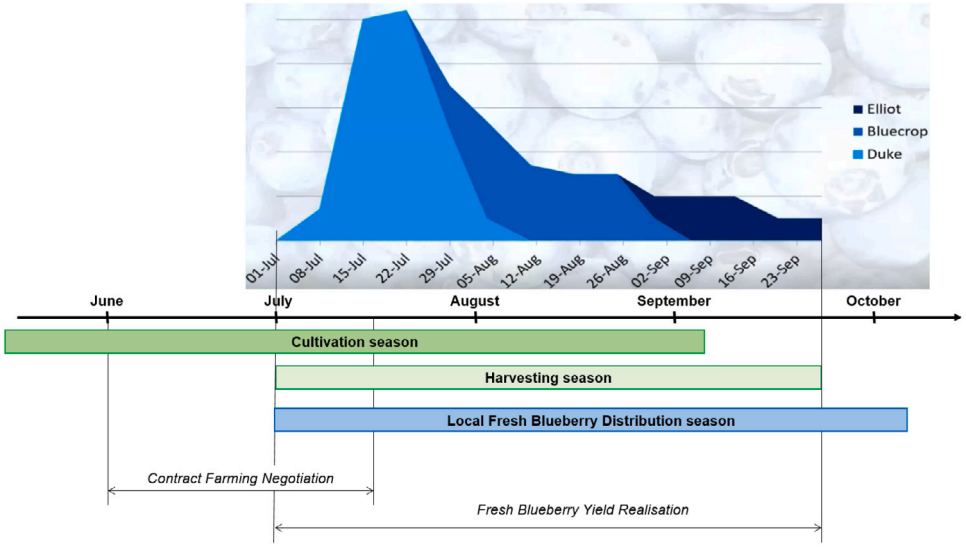


Fig. 2. The sequence of events in contract farming for three blueberry cultivars—'Elliott', 'Bluecrop', and 'Duke'—where different cultivars follow distinct ripening waves. *Source:* British Columbia Blueberry Council.

on pre-harvesting contract negotiation. Therefore, we focus on the strategic relationship between contracting and harvesting decisions, rather than on intra-seasonal price fluctuations.

Our study compares two types of contract farming: (1) penalty-based contract farming (traditional contract), and (2) incentive-based contract farming. Notably, the penalty-based agreements often reflect the industry's power dynamics, typically favouring distributors and potentially leading to less favourable terms for farmers. It is crucial to strive for equitable and viable contract farming, which ensures fair and balanced agreements between farmers and distributors for the continued prosperity of the sector.

4. Comparative analysis of penalty- and incentive-based contract farming

In this section, we conduct a thorough analytical comparison of the two types of contract farming, using the no-contract farming situation—where the farmer sells fresh produce directly to the spot market—as a baseline for our analysis. Accordingly, we develop models for three scenarios: no-contract farming, penalty-based contract farming, and incentive-based contract farming. The game sequences for each scenario are illustrated in Fig. 3. The proofs for all propositions are provided in Appendix.

4.1. No-contract farming

In this situation, the farmer determines the quantity of fresh blueberries Q_f^s to sell in the spot market. To model the problem analytically, we consider the two possible cases: (1) $Q_f^s > \tilde{Y}_f$, and (2) $Q_f^s \le \tilde{Y}_f$. To formulate the farmer's profit function considering these two cases, we develop the following model.

$$\Pi_F(Q_f^s) = \begin{cases} \tilde{p}_f^s \tilde{Y}_f + (p_z - c_z)(K\tilde{\xi} - \tilde{Y}_f) - (c_f + c_s)Q_f^s - C_s & if\ Q_f^s > \tilde{Y}_f \\ \tilde{p}_f^s Q_f^s + (p_z - c_z)(K\tilde{\xi} - Q_f^s) - (c_f + c_s)Q_f^s - C_s & if\ Q_f^s \le \tilde{Y}_f \end{cases} \tag{1}$$

where the realized fresh blueberry production that meets the quality requirements for the fresh blueberry market is expressed as $\tilde{Y}_f = K\tilde{\xi}\big(1 - H(\Theta_f)\big)$. In case 1, where the planned fresh blueberry quantity is greater than the realized high-quality fresh blueberry production, the first term in Eq. (1) represents the farmer's revenue obtained from selling fresh blueberries, which is a function of the uncertain spot market price and realized fresh blueberry yield. The second term shows the profit obtained from selling processed blueberries. The third term calculates the variable production and distribution costs of fresh blueberries. The last term denotes the fixed investment cost required to run the distribution system when selling directly to the spot market.

In case 2, where the planned fresh blueberry quantity is less than the realized fresh blueberry production, the farmer's revenue from selling the fresh and processed blueberries will differ from case 1. However, the variable production and fixed distribution costs remain the same as planned before the harvesting season.

Given $\tilde{Y}_f = K\tilde{\xi}\big(1 - H(\Theta_f)\big)$, the expected profit function of the farmer when selling directly to the market can be formulated as:

$$\begin{aligned} \mathbb{E}[\Pi_F(Q_f^s)] = {} & (\mathbb{E}[\tilde{p}_f^s] - c_f - c_s)Q_f^s + (p_z - c_z)(K\mathbb{E}[\tilde{\xi}] - Q_f^s) \\ & + (\mathbb{E}[\tilde{p}_f^s] - p_z + c_z) \\ & \times \left[\int_0^{\frac{Q_f^s}{K(1-H(\Theta_f))}} \Big(K\tilde{\xi}\big[1 - H(\Theta_f)\big] - Q_f^s\Big) g(\xi) d\xi\right] - C_s \end{aligned} \tag{2}$$

Proposition 1. *The optimal quantity of fresh blueberry Q_f^{*s} to be sold on the spot market by the farmer is:*

$$Q_f^{*s} = K\big[1 - H(\Theta_f)\big] G^{-1}\left(\frac{\mathbb{E}[\tilde{p}_f^s] - c_s - p_z + c_z - c_f}{\mathbb{E}[\tilde{p}_f^s] - p_z + c_z}\right) \tag{3}$$

4.2. Penalty-based contract farming

The distributor and farmer can enter into a traditional penalty-based contract farming arrangement (Chen and Chen, 2021). In the penalty-based contract farming model, farmers agree to supply distributors with a predetermined quantity of high-quality fresh blueberries on a future date in exchange for a guaranteed wholesale price. However, distributors typically have greater bargaining power, allowing them to dictate the penalty per unit of unfulfilled fresh blueberries (based on market and business norms), and also the wholesale price as price-makers, while farmers often take on the role of price-takers. Committing quantities beyond the expected yield might result in high penalties for farmers. Farmers retain some flexibility in their sales strategy, where they can choose to sell a portion of their blueberries as fresh through contracts while selling the remainder as processed goods on the spot market.

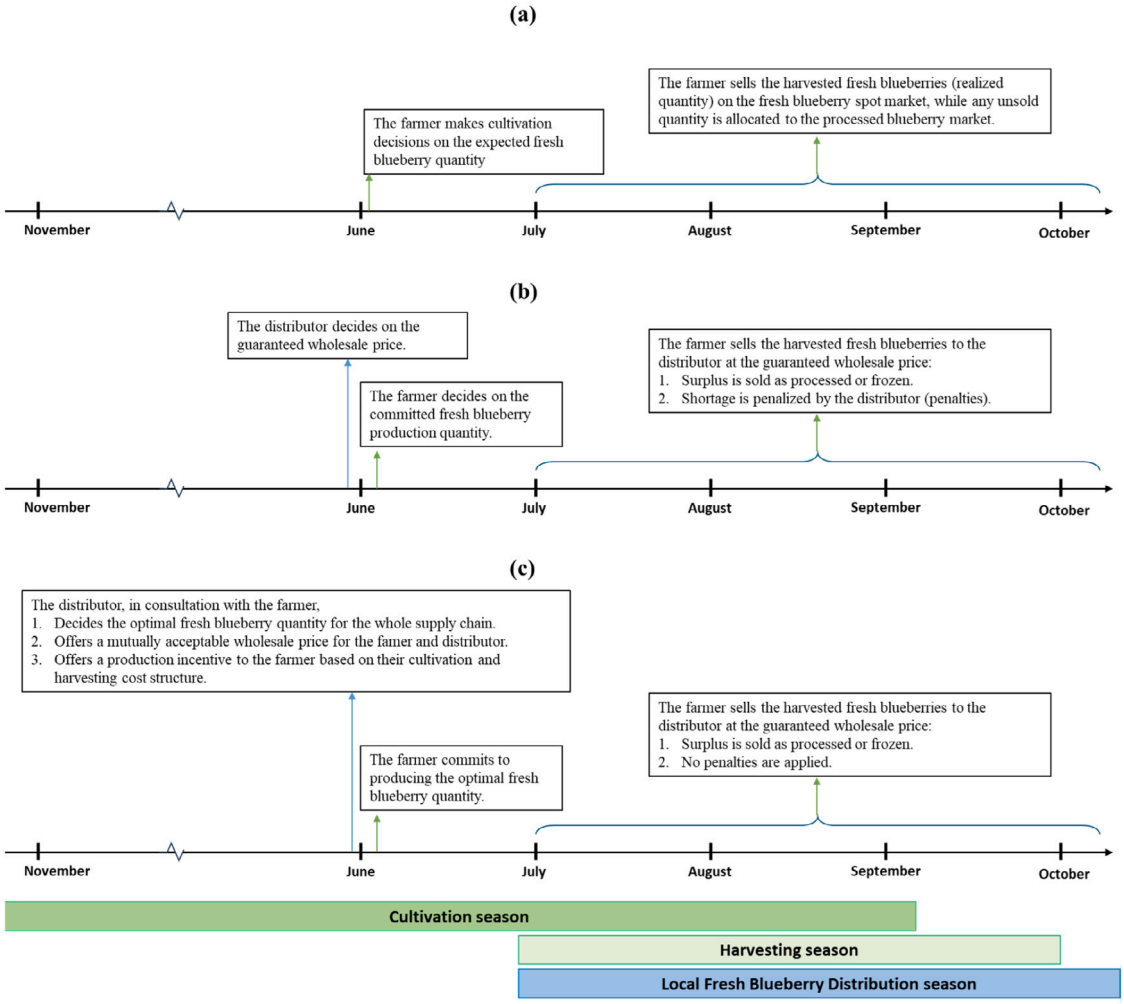


Fig. 3. The game sequences for (a) no-contract farming, (b) penalty-based contract farming, and (c) incentive-based contract farming scenarios.

Under the penalty-based contract farming model for blueberries, the farmer and the distributor make decisions before the harvesting season to determine the terms of their agreement. This situation is modelled as a Stackelberg leader–follower game. The distributor (leader) offers a guaranteed wholesale price (p_f^c) to the farmer for each unit of fresh blueberry supply, considering the future spot market price ($\tilde{p}_f^s$). The farmer (follower) then decides on a committed quantity of fresh blueberries (Q_f^c) to supply to the distributor during the harvesting season, based on factors such as the offered wholesale price (p_f^c), uncertain blueberry yield ($\tilde{\xi}$), penalty rate (π), and uncertain blueberry quality ($\tilde{\theta_f}$).

In the following, we use the backward induction method (Kicsiny et al., 2014) to solve the problem by obtaining the farmer's best-response function and then optimizing the distributor's problem based on the farmer's best-response function.

During the harvesting season, the farmer's actual yield of fresh blueberries ($\tilde{Y}_f$) may be lower or higher than the committed quantity (Q_f^c). The farmer may incur penalties for each unmet blueberry unit if the yield falls short of the contract volume. Including penalties in traditional contract farming agreements reflects real-world practices (Huh and Lall, 2013). On the other hand, if the actual yield exceeds the contract volume, the farmer may sell the excess amount on the processed blueberry market, as establishing a direct selling channel to sell fresh blueberries requires substantial advance planning. Considering the two cases (1) $Q_f^c > \tilde{Y}_f$, and (2) $Q_f^c \le \tilde{Y}_f$, the combined farmer's profit functions can be formulated as follows:

$$\Pi_F(Q_f^c) = p_f^c(Q_f^c \wedge \tilde{Y}_f) + (p_z - c_z)[K\tilde{\xi} - (Q_f^c \wedge \tilde{Y}_f)] - c_f Q_f^c - \pi p_f^c [Q_f^c - \tilde{Y}_f]^+ \tag{4}$$

where $(a \wedge b) = \min\{a, b\}$, and $[x]^+ = \max\{0, x\}$. Here, the first term represents the revenue received by the farmer from the distributor for the fresh blueberries delivered. The distributor will purchase either the committed quantity set in the contract or the farmer's actual yield of fresh blueberries, whichever is lower, taking into account quality and yield fluctuations. The second term denotes the profit obtained by the farmer by selling processed blueberries on the spot market. The third term shows the variable cost associated with fresh blueberry production. Finally, the last term represents the total penalty paid by the farmer to the distributor for the unmet fresh blueberry quantity.

Proposition 2. *The optimal quantity of fresh blueberry Q_f^{*c} to be committed by the farmer for any guaranteed wholesale price p_f^c offered by the distributor under penalty-based contract farming is as follows:*

$$Q_f^{*c} = K\left[(1 - H(\Theta_f))\right] G^{-1}\left(\frac{p_f^c - p_z + c_z - c_f}{p_f^c - p_z + c_z + \pi p_f^c}\right) \tag{5}$$

where $G(.)$ and $H(.)$ represent the CDF of the yield and quality random variables, respectively.

Although fresh and processed blueberry channels operate independently, they are inherently linked through farmers' allocation decisions.

The price p_z and cost c_z of processed blueberries influence how farmers allocate their harvest between the two channels considering their comparative profits. This interdependence is particularly evident in Proposition 2, where the processed blueberry price and cost directly affect the optimal commitment of fresh blueberries Q_f^{*c}. Consequently, the processed market serves not only as an alternative sales channel but also as a reference that shapes fresh market participation and contractual agreements.

Proposition 3. *The optimal amount of fresh blueberry committed by the farmer Q_f^{*c} under penalty-based contract farming rises as the guaranteed wholesale price offered by the distributor p_f^c rises. Besides, as p_f^c approaches maximum values, Q_f^{*c} tends to a threshold given by:*

$$G^{-1}\Big(\frac{1}{1+\pi}\Big)K\Big[1-H(\Theta_f)\Big] \tag{6}$$

Proposition 3 demonstrates the significant impact of the penalty coefficient and blueberry quality in reducing fresh blueberry production in contract farming. This finding sheds light on the relationship between the guaranteed wholesale price and the committed quantity, and it offers valuable insights into the behaviour of the optimal commitment as p_f^c increases.

During the harvesting season, the spot market price for fresh blueberries often exceeds the pre-harvest contract farming price. As a leader, the distributor determines the guaranteed wholesale price p_f^c offered to the farmer under penalty-based contract farming, considering the expected spot market price and the farmer's best-response strategy on the committed quantity. The distributor's objective function can be expressed as:

$$\begin{aligned}\Pi_D(p_f^c) = (\tilde{p}_f^s - p_f^c - c_d)\Big[Q_f^c \wedge K\tilde{\xi}\big[1-H(\Theta_f)\big]\Big] \\ + \pi p_f^c\Big[Q_f^c - K\tilde{\xi}\big[1-H(\Theta_f)\big]\Big]^+ - C_d\end{aligned} \tag{7}$$

The first term calculates the revenue obtained from selling the fresh blueberries minus the purchase and variable distribution costs. The second and third terms denote the penalty paid by the farmer and fixed distribution cost, respectively.

Proposition 4. *The optimal guaranteed wholesale price p_f^{*c} that the distributor should offer under the penalty-based contract farming satisfies:*

$$\frac{\Big(\pi(p_z-c_z+c_f)+c_f\Big)\Big(\mathbb{E}[\tilde{p}_f^s]-p_f^{*c}-c_d-\int_0^{\Omega}g(\xi)d\xi\big(\mathbb{E}[\tilde{p}_f^s]-p_f^{*c}-c_d-\pi p_f^{*c}\big)\Big)}{g(\Omega)\big(\Omega+(1+\pi)\int_0^{\Omega}(\xi-\Omega)g(\xi)d\xi\big)(p_f^{*c}-p_z+c_z+\pi p_f^{*c})^2} = 1 \tag{8}$$

where $\Omega = G^{-1}\Big(\frac{p_f^{*c}-p_z+c_z-c_f}{p_f^{*c}-p_z+c_z+\pi p_f^{*c}}\Big)$.

Proposition 5. *Under the penalty-based contract farming, the overall profit of the blueberry supply chain decreases as the penalty coefficient π increases.*

This result demonstrates the negative impact of penalty-based mechanisms on the profitability of the entire supply chain. While individual distributors may gain from imposing higher penalties on farmers who fail to meet their contractual commitments, this finding indicate that such measures ultimately undermine overall supply chain performance. To address these inefficiencies, the following section introduces an incentive-based contract farming model, which is designed to mitigate the adverse effects of penalties and promote a more collaborative and sustained contractual framework.

4.3. Incentive-based contract farming

In this section, we introduce a novel win–win incentive-based contract farming aimed at maximizing the profit of the entire supply chain while persuading both the farmer and the distributor to accept the new contract. The proposed incentive-based contract farming model is designed for a collaborative supply chain, emphasizing the importance of information sharing between the members of the supply chain. Both parties agree to share supply and demand information, fostering supply chain transparency and collaborative planning and forecasting.

In the incentive-based contract farming, the distributor refrains from penalizing the farmer for any shortfall in blueberry quantity and instead offers to incentive the farmer by covering a portion of the production costs. The incentive, denoted as α^{sc}, represents the proportion of the production unit cost that is subsidized by the distributor. Accordingly, under this contract, the distributor decides on the incentive α^{sc}, in addition to the guaranteed wholesale price p_f^{sc}, before the harvesting season. These contract parameters should be designed to convince the farmer to apply the fresh blueberry production quantity that maximizes the whole supply chain performance while also resulting in mutual benefits for both the distributor and the farmer. The incentive-based contract farming is designed under information symmetry, assuming that the parties have complete knowledge of each other's revenue and cost parameters (Niu et al., 2016). Designing contracts under information symmetry, where both parties have access to the same information, is common in the contracting literature (Macho-Stadler and Pérez-Castrillo, 2001). Information sharing can be considered one of the key requirements for creating collaborative win–win contracts, where both parties benefit.

To design incentive-based contract farming, we follow two widely adopted steps from contracting literature (Ilaria and Pierpaolo, 2004; Chaharsooghi and Heydari, 2010). First, the collaborative contract must be effective, meaning it can identify the optimal or nearly optimal point for the entire supply chain. Second, the collaborative contract must be desirable, meaning it is acceptable to all supply chain members.

Following the first step, we determine the optimal blueberry production quantity from the perspective of the whole supply chain to ensure the effectiveness of the incentive-based contract farming (Proposition 6). Subsequently, we design the contract parameters (Proposition 10) and establish analytical conditions (Propositions 12 and 13) to ensure the contract is effective and beneficial for both the farmer and the distributor.

Following the first step, we formulate the entire supply chain profit and determine the optimal decisions. The profit function of the entire supply chain can be expressed as:

$$\Pi_{SC}(Q_f^{sc}) = \begin{cases}(\tilde{p}_f^s - c_d)\tilde{Y}_f + (p_z - c_z)(K\tilde{\xi} - \tilde{Y}_f) - c_f Q_f^{sc} - C_d & if\ Q_f^{sc} > \tilde{Y}_f \\ (\tilde{p}_f^s - c_d)Q_f^{sc} + (p_z - c_z)(K\tilde{\xi} - Q_f^{sc}) - c_f Q_f^{sc} - C_d & if\ Q_f^{sc} \le \tilde{Y}_f\end{cases} \tag{9}$$

We now formulate Proposition 6 to establish the optimality condition on the fresh blueberry quantity for the whole supply chain.

Proposition 6. *The optimal quantity of fresh blueberry Q_f^{*sc} for the whole agri-food supply chain is given as follows:*

$$Q_f^{*sc} = K[1-H(\Theta_f)]G^{-1}\Big(1-\frac{c_f}{\mathbb{E}[\tilde{p}_f^s]-p_z-c_d+c_z}\Big) \tag{10}$$

Proposition 7. *The optimal quantity of fresh blueberry Q_f^{*sc} for the whole agri-food supply chain is always greater than (a) the optimal quantity of fresh blueberry to be sold on the spot market by the farmer Q_f^{*s}, and (b) the optimal quantity of fresh blueberry committed under the penalty-based contract farming Q_f^{*c}.*

Proposition 7 asserts that the optimal quantity of fresh blueberries Q_f^{*sc} for the entire agri-food supply chain invariably exceeds the optimal quantity designated for spot market sales Q_f^{*s} and committed under the penalty-based contract farming Q_f^{*c} by the farmer. The literature already establishes that the optimal production quantity for farmers, when viewed from the perspective of the entire supply chain, surpasses that obtained through traditional contract farming (Niu et al., 2016). Expanding on this result, Proposition 7 further demonstrates that the supply chain-optimal production quantity also exceeds that observed in direct spot market transactions without contractual agreements. This finding underscores the critical role of distributors in reducing logistics costs and enhancing overall supply chain efficiency, ultimately driving higher collective value creation.

Proposition 8. *A blueberry farmer will opt to sell fresh blueberries on the spot market if the anticipated market price exceeds a defined threshold. However, the expected price threshold at which the farmer and distributor agree to enter an incentive-based contract is lower than the threshold for selling independently on the spot market.*

These findings emphasize the pivotal role of distributors in enhancing the sale of fresh blueberries for farmers. Although literature frequently discusses the negative impacts of intermediaries on supply chains, such as the double-marginalization effect (Cachon, 2003), our results indicate that distributors can significantly support farmers by utilizing efficient logistics systems. This efficiency reduces the distribution costs for fresh blueberries, which is vital given the perishability of the product and the need for timely distribution.

To achieve the maximum performance of the supply chain, the farmer under collaborative contract farming must decide similar to Eq. (10). However, the farmer under traditional penalty-based contract farming determines Q_f^{*c} as $K(1-H(\Theta_f))G^{-1}\left([p_f^c - p_z + c_z - c_f]/[p_f^c - p_z + c_z + \pi p_f^c]\right)$, which is obtained in Proposition 2.

Proposition 9. *In the dual-channel blueberry supply chain, the distributor's decision to forfeit penalties is not a sufficient condition to achieve channel coordination.*

Proposition 9 highlights that while removing penalties in contract farming may mitigate some inefficiencies, it does not, on its own, ensure effective channel coordination. The absence of penalties eliminates the risk burden on farmers, but without an alternative mechanism to align incentives, the contract farming model ceases to provide sufficient economic advantages for the distributor. To address this, we introduce an incentive-based contract farming mechanism.

To incentivize the farmer to apply the optimal value of the whole channel (i.e., Eq. (10)), under the proposed incentive-based contract farming, the distributor sets the wholesale price and offers to pay for α^{sc} percent of the farmer's variable fresh blueberry production costs.

In the following, we obtain the conditions under which the farmer and distributor benefit from incentive-based contract farming. The farmer's profit function under incentive-based contract farming is given by Eq. (11), while the distributor's profit function under the new contract is given by Eq. (12).

$$\Pi_f(Q_f^{sc}) = p_f^{sc}(Q_f^{sc} \wedge \tilde{Y}_f) + (p_z - c_z)\left[K\tilde{\xi} - (Q_f^{sc} \wedge \tilde{Y}_f)\right] - (1-\alpha^{sc})c_f Q_f^{sc} \quad (11)$$

$$\Pi_D = (\tilde{p}_f^s - p_f^{sc} - c_d)\left[Q_f^{sc} \wedge K\tilde{\xi}[1-H(\Theta_f)]\right] - C_d - \alpha^{sc} c_f Q_f^{sc} \quad (12)$$

We now formulate Proposition 10 to establish the optimality condition for the fresh blueberry quantity under an incentive-based contract farming.

Proposition 10. *The fresh blueberry production quantity applied by the farmer is equal to the optimal supply chain production volume under all values of the guaranteed wholesale price p_f^{*sc} and incentive coefficient α^{*sc} that satisfy the following optimally condition:*

$$p_f^{*sc} = \mathbb{E}[\tilde{p}_f^s] - c_d - \alpha^{*sc}(\mathbb{E}[\tilde{p}_f^s] - p_z - c_d + c_z) \quad (13)$$

Providing the farmer with p_f^{*sc} and α^{*sc} will ensure that the planned fresh blueberry quantity is set to Q_f^{*sc}, leading to the maximum profit for the entire supply chain. However, varying p_f^{*sc} and α^{*sc} will result in different profits for the farmer and distributor. In the following, we will first analytically analyse the effects of various parameters on the incentive coefficient in Proposition 11. Then, we will determine the feasible range for the guaranteed wholesale price and incentive coefficient that ensures the participation of both parties.

Proposition 11. *The incentive coefficient (α^{*sc}) offered by the distributor decreases as either the unit variable distribution cost (c_d) for the distributor or the harvesting cost of processed blueberries (c_z) for the farmer increases. Conversely, an increase in the expected spot market price for fresh blueberries ($\mathbb{E}[\tilde{p}_f^s]$) or the selling price of processed blueberries (p_z) results in an increase in the incentive coefficient (α^{*sc}).*

A rise in the distributor's distribution costs would likely diminish their propensity to market fresh blueberries, consequently reducing the incentives offered to farmers. On the other hand, an increase in the anticipated price of fresh blueberries would prompt the distributor to enhance incentives, encouraging farmers to increase production to leverage the higher prices. Regarding the economics of processed versus fresh blueberries, a rise in the selling price of processed blueberries or a decrease in their variable costs necessitates an adjustment in the incentives. The distributor needs to boost incentives to maintain farmers' interest in cultivating fresh blueberries, ensuring farmers refrain from favouring the potentially more lucrative processed blueberry market.

To achieve win–win channel coordination, both members should be better off by accepting the contract. Accordingly, the farmer's profit under the incentive-based contract must be more than when they directly sells the fresh blueberry products to the spot market (i.e., situation 1). Additionally, the distributor's profit under an incentive-based contract must be more than penalty-based contract farming (i.e., situation 2).

We now formulate Proposition 12 to ensure the farmer's participation in the incentive-based contract farming.

Proposition 12. *The farmer will accept the incentive-based contract farming if $\mathbb{E}\left[\Pi_F(Q_f^{*sc}, p_f^{*sc}, \alpha^{*sc})\right] \geq \mathbb{E}\left[\Pi_F(Q_f^{*s})\right]$, which is hold when $p_f^{*sc} = \mathbb{E}[\tilde{p}_f^s] - c_d - \alpha^{*sc}(\mathbb{E}[\tilde{p}_f^s] - p_z - c_d + c_z)$ and $p_f^{*sc} \geq Max\{p_z - c_z, p_z - c_z + \Gamma\}$, where, (see Eq. (14) given as in Box I).*

The farmer's acceptance of incentive-based contract farming depends on whether the expected profit under the new contract is higher than that under the spot market. Specifically, the condition for the farmer's acceptance is $\mathbb{E}\left[\Pi_F(Q_f^{*sc}, p_f^{*sc}, \alpha^{*sc})\right] \geq \mathbb{E}\left[\Pi_F(Q_f^{*s})\right]$. To guarantee the farmer's participation, the wholesale price offered by the distributor should satisfy $p_f^{*sc} = \mathbb{E}[\tilde{p}_f^s] - c_d - \alpha^{*sc}(\mathbb{E}[\tilde{p}_f^s] - p_z - c_d + c_z)$, where $\mathbb{E}[\tilde{p}_f^s]$ is the expected spot market price. Furthermore, to ensure that the farmer does not face a loss compared to the spot market situation, the wholesale price should be greater than or equal to $Max\{p_z - c_z, p_z - c_z + \Gamma\}$, where Γ is given by Eq. (14).

The distributor offers a wholesale price in the feasible range and a suitable incentive coefficient to persuade the farmer to accept incentive-based contract farming. To accept the incentive-based contract farming model, the distributor requires a profit greater than the outcomes under traditional contract farming. This results in a maximum value of p_f^{*sc} that the distributor can offer without incurring a loss compared to traditional contract farming. We now formulate Proposition 13.

Proposition 13. *The distributor will apply the incentive-based contract farming if $\mathbb{E}\left[\Pi_D(Q_f^{*sc}, p_f^{*sc}, \alpha^{*sc})\right] \geq \mathbb{E}\left[\Pi_D(Q_f^{*c}, p_f^{*c})\right]$, which is hold when*

$$\Gamma = \frac{(\mathbb{E}[\tilde{p}_f^s] - p_z - c_d + c_z)\left(\mathbb{E}\left[\Pi_F(Q_f^{*s})\right] - K\mathbb{E}[\xi](p_z - c_z)\right)}{(\mathbb{E}[\tilde{p}_f^s] - p_z - c_d + c_z - c_f)Q_f^{*sc} - (\mathbb{E}[\tilde{p}_f^s] - p_z - c_d + c_z)\int_0^{\frac{Q_f^{*sc}}{K[1-H(\Theta_f)]}}(Q_f^c - K\xi\left[1 - H(\Theta_f)\right])g(\xi)d\xi} \tag{14}$$

Box I.

$$\Psi = \frac{(\mathbb{E}[\tilde{p}_f^s] - p_z - c_d + c_z)\left(\mathbb{E}\left[\Pi_D(Q_f^{*c}, p_f^{*c})\right] + C_d\right)}{(\mathbb{E}[\tilde{p}_f^s] - p_z - c_d + c_z)\left(\int_0^{\frac{Q_f^{*sc}}{K\left[1-H(\Theta_f)\right]}}\left(K\xi\left[1 - H(\Theta_f)\right] - Q_f^{*sc}\right)g(\xi)d\xi + Q_f^{*sc}\right) - c_f Q_f^{*sc}} \tag{15}$$

Box II.

$p_f^{*sc} = \mathbb{E}[\tilde{p}_f^s] - c_d - \alpha^{*sc}(\mathbb{E}[\tilde{p}_f^s] - p_z - c_d + c_z)$ *and* $p_f^{*sc} \le Min\left\{\mathbb{E}[\tilde{p}_f^s] - c_d, \mathbb{E}[\tilde{p}_f^s] - c_d - \Psi\right\}$*, where, (see Eq. (15) in Box II).*

To ensure that the incentive-based contract farming is of benefit to the distributor, the expected profit from the scheme must exceed the distributor's standard operations. This condition holds if the guaranteed wholesale price and incentive coefficient are such that the farmer's expected price is given by $\mathbb{E}[\tilde{p}_f^s] - c_d - \alpha^{*sc}(\mathbb{E}[\tilde{p}_f^s] - p_z - c_d + c_z)$ and the price is no higher than $\min\left(\mathbb{E}[\tilde{p}_f^s] - c_d, \mathbb{E}[\tilde{p}_f^s] - c_d - \Psi\right)$, where Ψ is given by Eq. (15). Here, $\mathbb{E}[\tilde{p}_f^s]$ is the expected price of fresh blueberries, c_d is the distributor's fixed cost per unit of blueberries, p_z and c_z are the market price and production cost of processed blueberries per unit, respectively, and α^{*sc} is the incentive coefficient. Q_f^{*c} and p_f^{*c} are the optimal fresh blueberry quantity and price for the distributor's standard operations, and $\mathbb{E}\left[\Pi_D(Q_f^{*c}, p_f^{*c})\right]$ is the expected profit from those operations. The remaining terms have their previously defined meanings.

Proposition 14. *In the dual-channel blueberry supply chain, the total supply chain profit achieved through incentive-based contract farming is always higher than that under penalty-based contract farming.*

Proposition 14 demonstrates that incentive-based contract farming will always yield higher supply chain surplus than penalty-based contract farming. This important finding should promote coordination and collaboration between farmers and distributors. Although penalty-based contract farmings have been widely applied in practice (UNIDROIT, 2015), our finding indicates that an incentive-based contract would be more efficient in improving supply chain performance. Our findings align with observations in behavioural economics. To the best of our knowledge, this study is the first to analytically establish this result in the contract farming literature.

So far, in this study, we focused on a dual-channel blueberry supply chain. Nonetheless, our approach can also be extended to a single-channel supply chain, where farmers are restricted to selling only fresh blueberries. To do so, we set the processed blueberry profit margin to zero (i.e., $p_z - c_z = 0$), effectively removing the processed market from the decision-making process. In the following, we introduce new propositions to analytically examine how the transition from a dual-channel to a single-channel structure affects the incentive coefficient and guaranteed wholesale price, providing insights into the economic implications of each market configuration.

Proposition 15. *(a) The incentive coefficient α^{*sc} is higher in a dual-channel supply chain compared to a single-channel fresh blueberry supply chain; (b) If the incentive coefficient is constant, then the guaranteed wholesale price p_f^{*sc} in the dual-channel supply chain will exceed that of the single-channel fresh supply chain; (c) If the profit margin of processed blueberries $(p_z - c_z)$ increases, then both the incentive coefficient and the guaranteed wholesale price offered by the distributor will also increase.*

Proposition 15 underscores the advantages of dual-channel supply chain for farmers. Indeed, having the option to sell both fresh and processed blueberries, farmers enhance their bargaining power in contract negotiations with distributors. Consequently, to encourage farmers to participate in contract farming, distributors may need to offer higher incentive coefficients or increase the guaranteed wholesale price. Moreover, as the profit margin of processed blueberries increases, the distributor is likely to adjust the incentive structure accordingly, further improving contract terms for farmers. From a practical perspective, these results suggest that a dual-channel distribution model can serve as a mechanism to mitigate power imbalances in contract farming negotiations, fostering more equitable and sustained supply chain relationships.

5. Numerical experiments

We conducted a numerical analysis to evaluate the effectiveness of the proposed contract farming strategies. We adjusted model parameters using data from the blueberry industry in British Columbia and other reliable sources. Several numerical simulations and sensitivity tests were performed to validate our theoretical results.

5.1. Data

For our numerical investigations, we utilized data from the USDA, field data from blueberry farms in British Columbia, and recent academic research. We focused on a typical blueberry farmer in the Fraser Valley, operating on a 40-acre farm, growing blueberries for fresh and processed markets.

To assess the uncertainty of fresh blueberry market prices, we analysed USDA weekly price data (2000–2020), fitting it to a normal distribution with a mean price of $\mathbb{E}[\tilde{p}_f^s] = 2.32\$$ and a standard deviation of $\sigma_{\tilde{p}_f^s} = \$0.62$. Processed blueberry prices are more stable, with a market price of \$0.95/lb and production/harvesting costs of \$0.55/lb and \$0.37/lb for fresh and processed blueberries, respectively (Galinato et al., 2016; US Highbush Blueberry Council, 2020).

Yield uncertainty was modelled for North American blueberry acreages, assuming a normal distribution with a mean yield of $\mu_g = 15000$ lb/acre and a standard deviation of $\sigma_g = 4000$ lb (Washington State Fruit Commission, 2020). Quality standards require a minimum blueberry diameter of $\Theta_f = 8$ mm, which follows a normal distribution with a mean of $\mu_h = 7.92$ mm and a standard deviation of $\sigma_h = 0.69$ mm (Rowland et al., 2020; Ministry of Agriculture and Food, 2007). Based

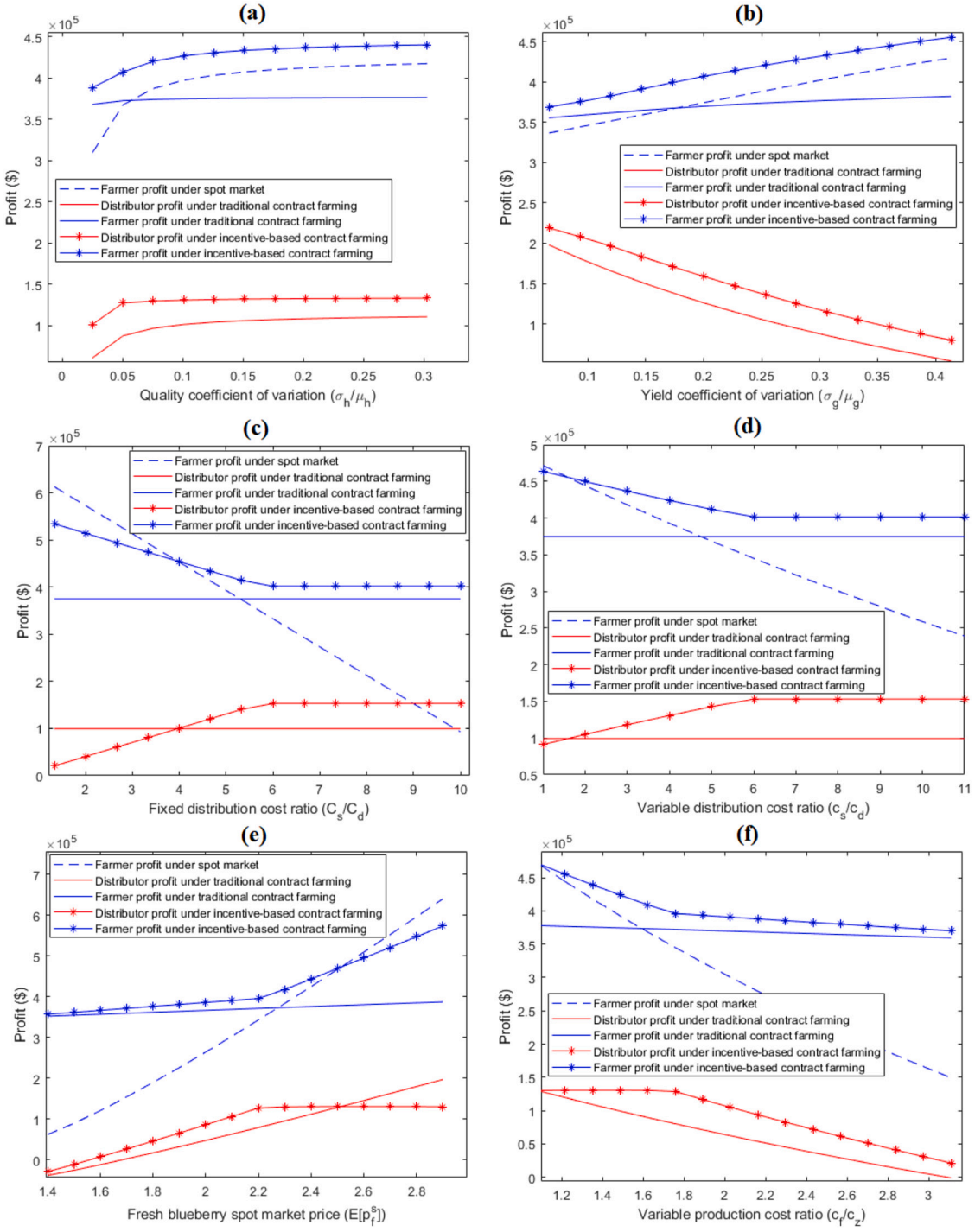


Fig. 4. Changes in the farmer and distributor profit in response to variations in (a) quality uncertainty, (b) yield uncertainty, (c) fixed distribution cost, (d) variable distribution cost, (e) market price, and (f) production/harvesting cost.

on these criteria, approximately 51% of the realized yield qualifies for the fresh market (Evans and Ballen, 2019).

In penalty-based contracts, farmers face a penalty coefficient of 150% ($\pi = 1.5$) for unfulfilled orders, aligning with standard contract farming practices (Huh and Lall, 2013). Farmers incur fixed and variable distribution costs of $C_s = 300{,}000$ and $c_s = 0.2$, respectively, while the distributor's fixed and variable distribution costs are $C_d = 60{,}000$ and $c_d = 0.05$.

5.2. Numerical results

We applied these parameters to derive the decision variables and profits, evaluating the performance across three decision-making scenarios. The results are presented in Table 2.

Table 2 demonstrates the exciting potential of the incentive-based contract farming scenario. It yields the highest fresh produce quantity

(Q_f), indicating a more significant potential for selling fresh blueberries in the market under this approach. In penalty-based contract farming, the distributor leverages their market power as the dominant party. However, transitioning to an incentive-based contract farming model results in significantly more significant benefits. This shift not only enhances the distributor's profits but also substantially improves the farmer's earnings and overall supply chain efficiency, leading to a thrilling increase in financial gains.

The incentive-based contract farming model not only ensures a win–win situation for both the distributor and the farmer, with profits surpassing those in the other scenarios, but also benefits the entire supply chain. By increasing overall profits by 17% compared to the penalty-based contract and by 41.2% compared to the no-contract farming scenario, it demonstrates its potential for widespread positive impact, reinforcing its effectiveness and fairness.

The proposed incentive-based contract farming approach, which involves optimizing adjustments to the wholesale price and fairly distributing production and harvesting costs, not only leads to improved outcomes for both stakeholders but also plays a crucial role in enhancing the efficiency of the entire supply chain. This key factor instills confidence in the audience about the proposed approach's potential to significantly enhance the industry's productivity.

5.3. Sensitivity analyses

Our novel numerical investigations and sensitivity analyses bear significant implications for the agri-food industry. By rigorously testing the theoretical findings and evaluating the model's behaviour under diverse conditions, we sought to assess how modifications to yield and quality uncertainties, market price expectations, distribution costs, and production/harvesting costs impact the efficacy of the proposed contracts. These unique findings are pivotal for comprehending the potential influence of contract design on the industry's profitability and stability.

The outcomes of our robust sensitivity analysis, showcased in Figs. 4 and 5, vividly demonstrate the influence of variations in quality uncertainty, yield uncertainty, distribution costs, production costs, and market price on the feasibility and profitability of the proposed contracts. This comprehensive analysis offers a practical foundation for understanding the potential outcomes of different contract scenarios, thereby aiding industry professionals, researchers, and policymakers in making informed decisions.

Fig. 4a, b, and f suggest that the incentive-based contract is more efficient than both the no-contract and penalty-based contract scenarios in boosting profits for all members of the agri-food supply chain under fluctuations in quality, yield, and production cost ratios. Fig. 4c, d, and e indicate that the incentive-based contract outperforms the other scenarios under specific conditions, particularly when variations in parameters such as fixed distribution cost ratio, variable distribution cost ratio, and spot market price fall below or rise above certain thresholds. These significant observations call for further investigation to fully grasp their implications for the industry.

Figs. 5b and 4b illustrate the unwavering adaptability of the incentive-based contract model. It consistently yields higher profitability across varying levels of yield uncertainty, adjusting the wholesale price offered to farmers to maintain a stable feasible interval. Fig. 4a further demonstrates that quality uncertainty does not impact the viability of the incentive-based contract. Both farmers and distributors achieve higher profits even under high-quality uncertainty, a testament to the robustness of this contract model.

Figs. 5c and 5d reveal that the incentive-based contract remains viable when the farmer's distribution costs are significantly higher than those of the distributor. This finding aligns with real-world scenarios, such as when a farmer has to transport their produce from a remote location, or when a distributor has a well-established distribution network, where distributors leverage economies of scale, specialized knowledge, and extensive distribution networks to reduce costs. In contrast, when the farmer's distribution costs are low, direct market sales may become more advantageous.

Table 2
Decision variables and profits for the three decision-making situations.

	Situation 1: No contract farming	Situation 2: Penalty-based contract farming	Situation 3: Incentive-based contract farming
Q_f	284 994	166 399	305 228
p_f	n/a	1.328	1.062
α^{sc}	n/a	n/a	0.715
$E(\Pi_F)$	392 789	374 634	424 039
$E(\Pi_D)$	n/a	99 384	130 634
$E(\Pi_{SC})$	392 789	474 018	554 673

Fig. 5e suggests that while the incentive-based contract remains feasible under current market price expectations, it becomes challenging to implement if market prices rise significantly. In such cases, farmers demand higher wholesale prices, potentially making direct market sales more attractive. This dynamic underscores the crucial importance of maintaining realistic market price expectations in the contract design.

Fig. 5f confirms that the incentive-based contract remains beneficial even when production/harvesting costs, which include the cost of seeds, fertilizers, labour, and machinery, for fresh blueberries are high. The feasible wholesale price interval widens as these costs increase, demonstrating the model's robustness. This flexibility is crucial for accommodating varying production cost structures within the agri-food supply chain.

Fig. 6 depicts variations in the fresh blueberry production quantity across different situations resulting from changes in critical model parameters. As evident from various segments of Fig. 6, the amount committed by the farmer under the traditional penalty-based contract generally exceeds the quantity planned for the direct selling situation. This trend can be attributed to the guaranteed wholesale price offered by the distributor under a traditional contract, which mitigates market price risks for the farmer. In comparison, the fresh blueberry production quantity under an incentive-based contract surpasses that under a conventional penalty-based contract since an incentive-based contract mitigates market price risk and eliminates the risk of penalties, resulting in a higher amount of fresh blueberry quantity committed by the farmer.

Fig. 7 indicates that the incentive-based production-management contract farming consistently outperforms traditional penalty-based contract farming in terms of profitability for the entire channel. The proposed incentive-based contract farming outperforms the traditional penalty-based approach across different levels and scenarios of quality (Fig. 7a), yield (Fig. 7b), and price (Fig. 7e) uncertainty, as well as various combinations of cost parameters (Fig. 7c, d, and f). However, in a specific instance shown in Fig. 7c, if the farmer's cost of reaching the market through direct selling is low enough compared to the distributor's cost, it would be more profitable if the farmer sells directly to the market rather than participating in either incentive-based or traditional contract farming. Nevertheless, distributors benefit from economies of scale due to their specialized knowledge, equipment, and skills, enabling them to maintain considerably lower distribution costs than non-specialized farmers.

Fig. 8 provides valuable insights into the performance of incentive-based contract farming across different penalty coefficient levels. Fig. 8a and b demonstrate a positive correlation between increasing penalty coefficients and improvements in supply chain, farmer, and distributor outcomes under incentive-based contract farming. This highlights the detrimental impact of punitive measures inherent in traditional contract farming on supply chain performance.

Moreover, Fig. 8c interestingly shows that the applicability of incentive-based contract farming increases as the penalty coefficient

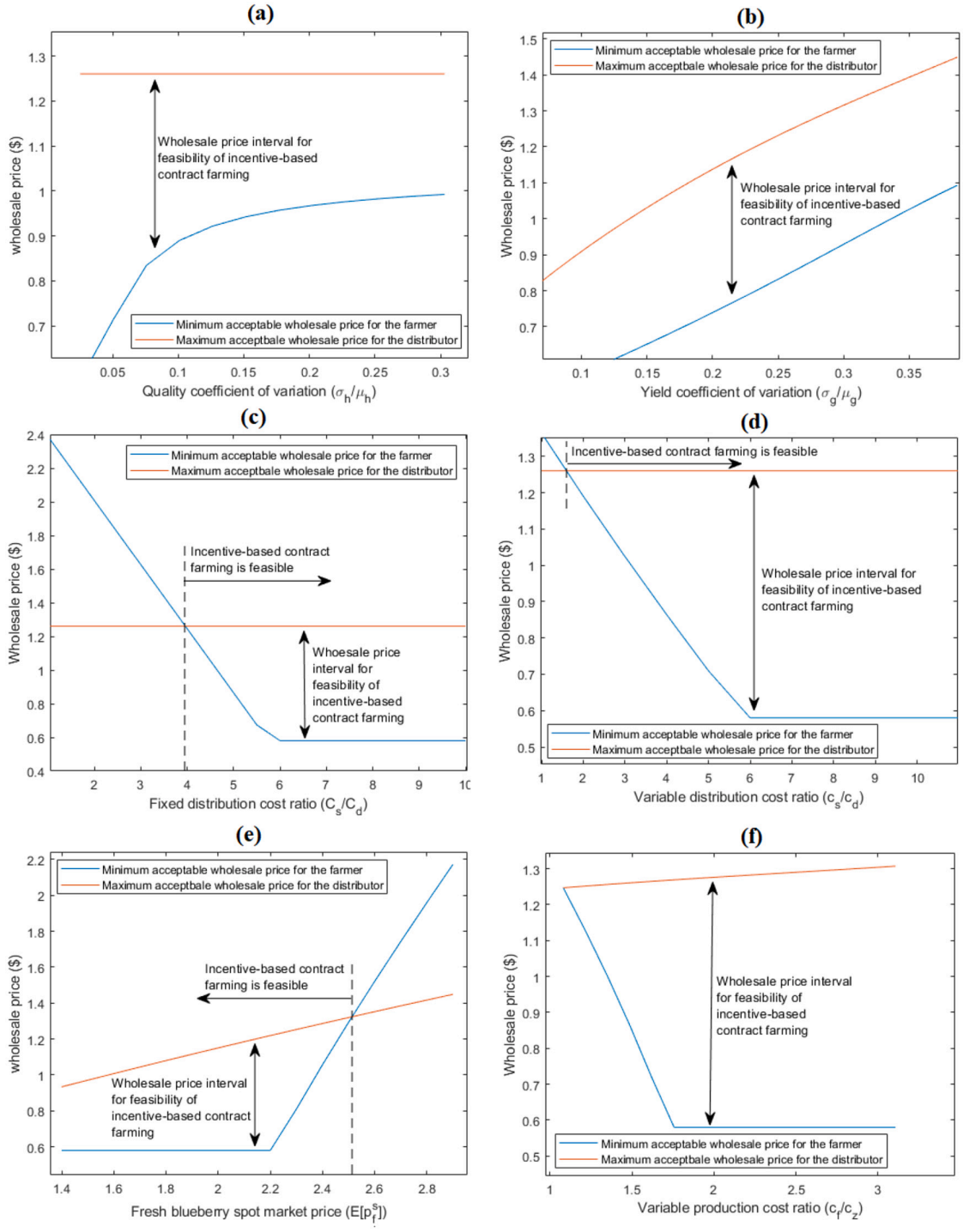


Fig. 5. Feasible wholesale price interval in response to variations in (a) quality uncertainty, (b) yield uncertainty, (c) fixed distribution cost, (d) variable distribution cost, (e) market price, and (f) production/harvesting cost.

rises. This significant finding emphasizes the flexibility of incentive-based models to adapt to various real-world conditions. Additionally, Fig. 8d reveals that while traditional contract farming's focus on penalty coefficients may discourage distributors from selling blueberries as fresh, incentive-based contract farming is crucial in enabling farmers to market a significant portion of their produce as fresh, thus supporting the fresh produce market effectively.

6. Managerial insights

Our study demonstrates the superiority of incentive-based contract farming under conditions of information symmetry. Through rigorous analytical and numerical analysis, we have shown that incentive-based contracts not only align the interests of farmers and distributors but also pave the way for higher profits for both parties compared to traditional penalty-based contracts.

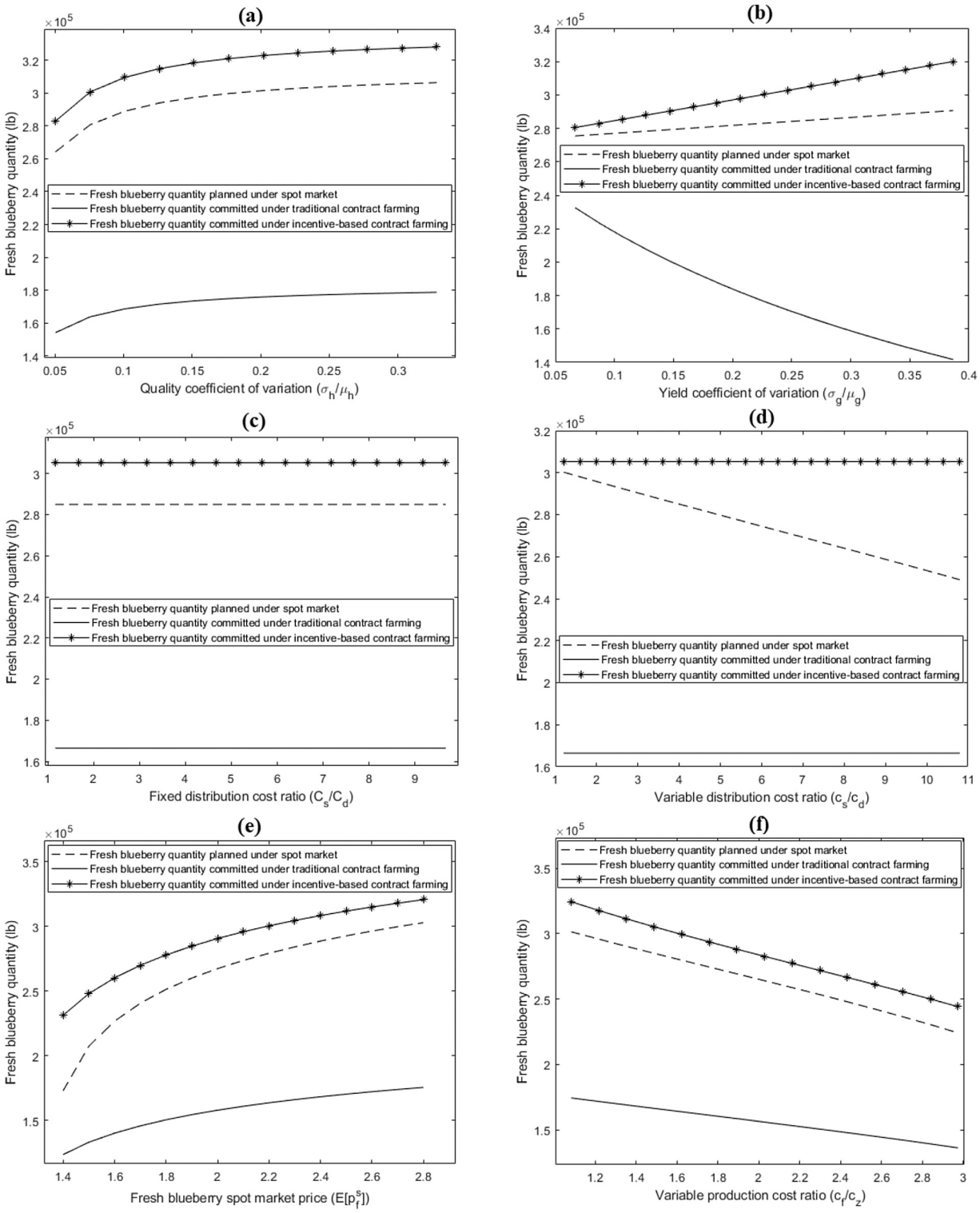


Fig. 6. Planned/committed fresh blueberry quantity in response to variations in (a) quality uncertainty, (b) yield uncertainty, (c) fixed distribution cost, (d) variable distribution cost, (e) market price, and (f) production/harvesting cost.

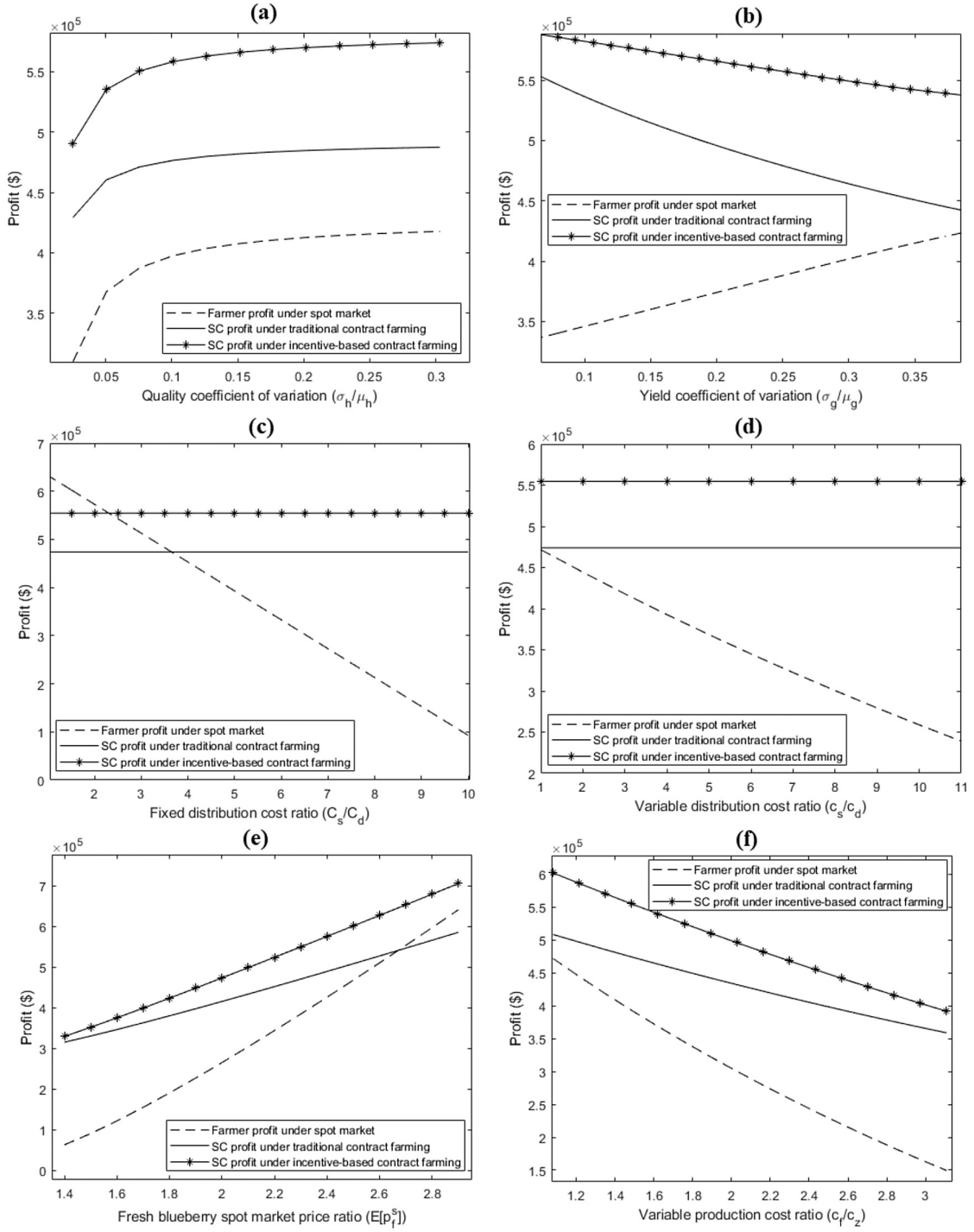


Fig. 7. Changes in supply chain profit in response to variations in (a) quality uncertainty, (b) yield uncertainty, (c) fixed distribution cost, (d) variable distribution cost, (e) market price, and (f) production/harvesting cost.

The introduction of production cost-sharing mechanisms within incentive-based contracts is critical for fostering a collaborative and equitable relationship between farmers and distributors. By sharing both the benefits, such as increased profits, and risks, such as crop failure or market price fluctuations, these contracts can transform traditional power dynamics into a partnership based on mutual trust and shared objectives. This approach incentivizes farmers to invest in quality improvements and aligns their efforts with the goals of distributors while securing revenues.

Supply chain managers should prioritize adopting incentive-based contracts over penalty-based models. These contracts mitigate risks associated with yield and quality uncertainties by providing a more flexible and collaborative framework. They motivate farmers to optimize production and ensure higher profits for farmers and distributors, enhancing overall supply chain performance.

Distributors should leverage their economies of scale and specialized distribution capabilities to make incentive-based contracts more attractive to farmers. Distributors can offer competitive terms that

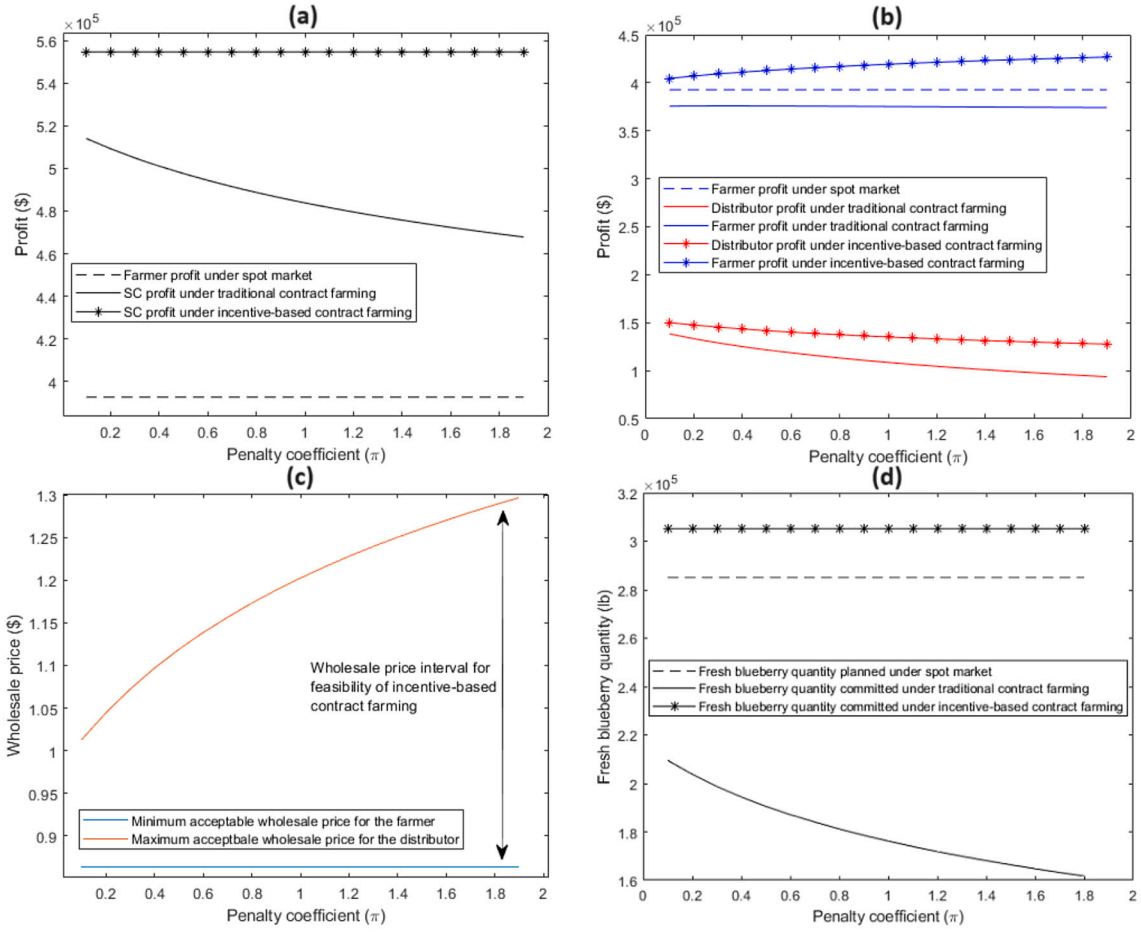


Fig. 8. Changes in supply chain profit, farmer and distributor profit, planned/committed fresh blueberry quantity, and feasible wholesale price interval in response to variations in penalty coefficient.

benefit both parties by investing in efficient distribution systems. This investment not only reduces costs but also improves the reliability and reach of the supply chain.

Farmers should consider incentive-based contracts to mitigate the risks associated with production cost. These contracts offer stable and predictable income, making them an attractive alternative to direct market sales, especially in volatile market conditions. By securing a guaranteed wholesale price, farmers can plan their production more effectively and reduce the financial risks of market price variability.

7. Conclusion

The literature has long advocated contract farming to enhance supply chain efficiency and productivity. However, there needs to be more analytical proof that such arrangements benefit farmers, as buyers often exploit their power. Our study introduces an innovative incentive-based contract farming model that not only increases profitability across the supply chain but also ensures fair benefits for both parties. By capturing the main features of blueberry farming, we demonstrate the effectiveness of the proposed incentive-based contract farming under various sources of uncertainty, especially when farmers face challenges in adjusting their production capacity.

Our findings underscore the importance of adopting incentive-based contracts within the agri-food supply chain. These contracts provide a robust and mutually beneficial alternative to traditional penalty-based models. By integrating production cost-sharing mechanisms and fostering a spirit of collaboration, incentive-based contracts enhance profitability, align the interests of all supply chain participants, and improve overall supply chain efficiency.

This research offers valuable insights into designing and implementing effective contract farming arrangements. By adopting these practical and proven strategies, agri-food supply chain managers can create a more collaborative and equitable supply chain that benefits all stakeholders. Importantly, our study provides practical guidance on how to implement incentive-based contracts, laying the groundwork for future research exploring the long-term impacts of these contracts and their potential to drive sustainable agricultural practices.

Future studies should consider a multi-period and dynamic approach to evaluate the long-term benefits of incentive-based contracting. For instance, we used a combination of historical data analysis and scenario modelling to assess the potential benefits of our model over a 5-year period. Comprehensive risk analyses using measures like Conditional Value-at-Risk should be conducted to assess potential risks for farmers and agribusiness firms. Additionally, while this study individually examines various sources of uncertainty, future research could integrate correlations between quality, yield, and price to provide a more comprehensive analysis. Finally, designing incentive-based contract farming models that account for information asymmetry between farmers and distributors could be a promising direction for future research.

CRediT authorship contribution statement

Mohammadreza Nematollahi: Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Adel Guitouni:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration,

Funding acquisition, Conceptualization. **Jafar Heydari:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **Eric M. Gerbrandt:** Validation, Data curation, Conceptualization.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijpe.2025.109635.

Data availability

Data will be made available on request.

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